

IT-Supported Request Management for Clinical Radiology: Contextual Design and Remote Prototype Testing

Philipp Krop*

Samantha Straka*

philipp.krop@uni-wuerzburg.de
samantha.straka@uni-wuerzburg.de
HCI Group, University of Würzburg
Würzburg, Bavaria, Germany

Maximilian Ertl

Service Center Medical Informatics, University Hospital of
Würzburg
Würzburg, Bavaria, Germany
ertl_m@ukw.de

Melanie Ullrich

Service Center Medical Informatics, University Hospital of
Würzburg
Würzburg, Bavaria, Germany
ullrich_m@ukw.de

Marc Erich Latoschik

HCI Group, University of Würzburg
Würzburg, Bavaria, Germany
marc.latoschik@uni-wuerzburg.de

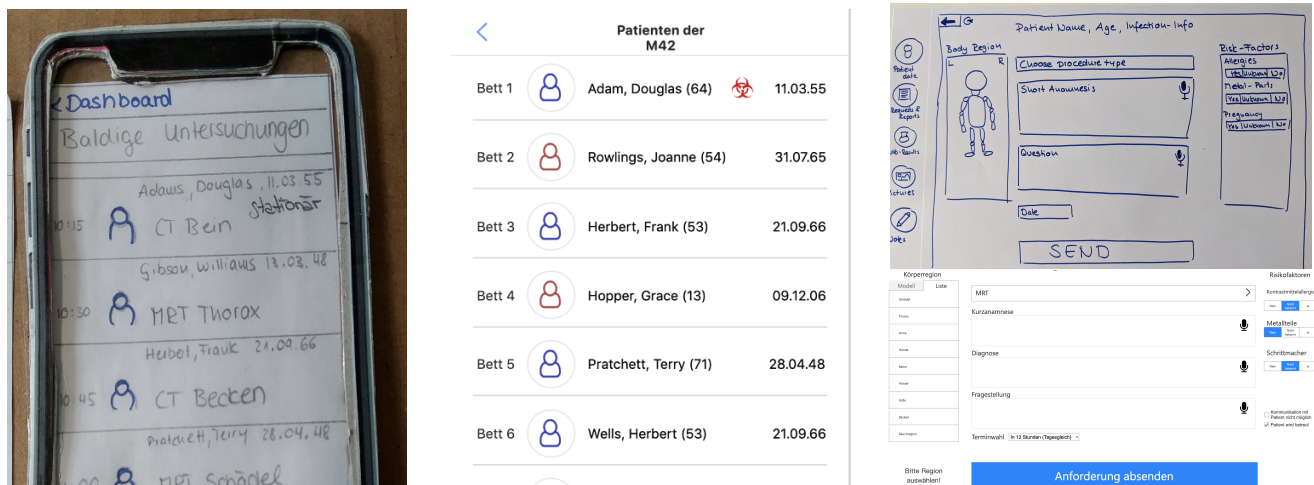


Figure 1: From Paper to Digital: The results of prototyping a mobile and a desktop app for radiological request management.

ABSTRACT

Management of radiology requests in larger clinical contexts is characterized by a complex and distributed workflow. In our partner hospital, representing many similar clinics, these processes often still rely on exchanging physical papers and forms, making patient or case data challenging to access. This often leads to phone calls with long waiting queues, which are time-inefficient and result in frequent interrupts. We report on a user-centered design approach based on Rapid Contextual Design with an additional focus group to optimize and iteratively develop a new workflow.

*Both authors contributed equally to this research.

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Participants found our prototypes *fast and intuitive*, the *design clean and consistent*, *relevant information easy to access*, and the *request process fast and easy*. Due to the COVID pandemic, we switched to remote prototype testing, which yielded equally good feedback and increased the participation rate. In the end, we propose best practices for remote prototype testing in hospitals with complex and distributed workflows.

CCS CONCEPTS

• **Human-centered computing** → Field studies; User centered design; Contextual design; • **Social and professional topics** → Health information exchanges.

KEYWORDS

Rapid Contextual Design, HCI, User-Centered Design, Prototyping, Remote Testing, Radiology

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1 INTRODUCTION

Although technology is commonly believed to provide great potential to improve or optimize workflows in nearly every aspect of life, German hospitals are severely behind in digitization [18]. In a survey by Rudolf and Schmidt, 80% of German medical doctors stated that digitization would improve their work [17]. However, a naive approach to digitization will not reveal the full potential of digital solutions since user resistance and bad usability are significant barriers for IT solutions in hospitals [6, 20].

This work is part of an overarching interdisciplinary project funded by the *Bavarian State Ministry for Science and the Arts*. It aims to improve workflows in hospitals through digitization using AI, user-centered design, and data mining. One critical workflow in need of improvement is the management of radiology requests, in which radiological examinations such as X-rays or magnetic resonance tomography are requested, conducted, and reported. Here, an optimization not only addresses aspects of efficiency and effectiveness but also affects primary health issues of the patients: An internal analysis of the respective request management process in the *University Clinic of Würzburg* revealed that, often, physicians expose patients to unnecessary radiation. Like other doctors worldwide, they underestimate harmful radiation dosages [2, 4], as they request multiple radiological examinations, even for small check-ups. Even though they use specialized software and tools, doctors are often unaware that a procedure has already been conducted for a patient. This could indicate several problems, including usability issues or a lack of communication between the physicians and the radiologists. To tackle these problems, we decided to analyze and improve this radiological request management process using the *Rapid Contextual Design* approach [5], which we believe is especially suited in this context of multiple involved parties because it relies on user involvement, thus increasing user acceptance. We already detailed the results of our requirement analysis in our prior work [11], where we found nine major problem areas, as summarized in table 1. This initial analysis is the foundation for improvements in workflow and user interfaces.

Contribution: We used Contextual Interviews and focus groups to analyze the needs and problems of users with different responsibilities in the radiological request management process, which we also detailed in our previous work [11]. From there, we restructured the workflow and iteratively developed mobile and desktop prototypes based on our findings. We conducted prototype tests, in which we assessed perceived usability, task load, and qualitative feedback. Additionally, we describe how we switched to remote testing in response to the COVID-19 pandemic and how it impacted our work. Our results indicate that contextual design is suitable for analyzing and improving processes in hospitals and that remote prototype testing may even be more feasible for hospitals than in-person testing. We also describe how we changed the Rapid

Contextual Design approach to accommodate constraints in the hospital setting. An additional focus group improved our understanding of the underlying processes, constraints, and interactions between end-users. At the end of this publication, we offer five recommendations for Rapid Contextual Design and remote prototype testing in hospitals. We believe these changes and guidelines to be interesting for others who strive to analyze and improve complex and distributed workflows in hospitals.

2 METHODS

Our overall goal is to improve the radiological request management process in our partner hospital, the *University Clinic of Würzburg*. Request management spans the whole process from requesting a specific radiological procedure up to the final report, which is used by the physician to treat a patient. It is a very complex and distributed workflow: five different roles (*physician, radiologist, medical assistant for scheduling, medical assistant for the procedure and supervising radiologist*) have to work together on time, often supported by various different independent software tools. This process can be seen on the left side of Figure 2. It consists of the following steps:

- (1) A *physician* writes a new request for a procedure using SAP.
- (2) A *radiologist* has to confirm its medical necessity and safety. If those are not given or unclear, the *radiologist* can reject the procedure and send it back to the *physician*.
- (3) Meanwhile, *medical assistants* in the radiology department schedule the procedure via phone with the *physician*.
- (4) The procedure is performed, supervised by *radiologists* and *medical assistants*.
- (5) When the images are finished, a *radiologist* looks at them and reports the findings, which a *supervising radiologist* then checks.
- (6) The final report is sent back to the *physician* who made the request.

There are many ways to approach user-centered design, such as focus groups [12], usability testing, and surveys [22, 23]. As we wanted to improve and understand the user interface of the current system and the whole process, it was necessary to use a holistic approach such as *Design for Well Being* [10] or *Rapid Contextual Design* [5], which have been used in similar settings before [19, 21]. We chose *Rapid Contextual Design* as it is already successfully used for distributed workflows [16], can be used in the field, and helps to understand processes rapidly without disturbing the target users. It consists of six phases: In the *Data Aggregation* phase, the researchers gather information about how daily work is performed, where the problems lie, and what the workarounds are. This is done with Contextual Interviews, where researchers accompany target users in the field. These interviews are then analyzed in the *Analysis* phase with a hierarchical sorted diagram. Based on these findings, the researchers develop *visions* and *storyboards* and validate them with the stakeholders. Visions and storyboards both graphically detail the new workflow with varying levels of detail, the lesser being the most detailed and including basic user interface sketches. Both visions and storyboards are evaluated with the stakeholders multiple times. Finally, an application *prototype* based on the final

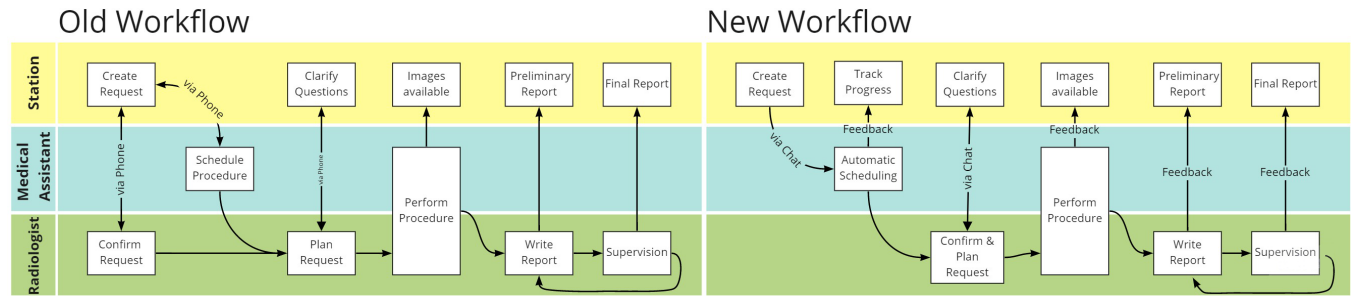


Figure 2: The old workflow (left) and our improved version of it (right). The most noticeable changes are better communication channels and feedback loops.

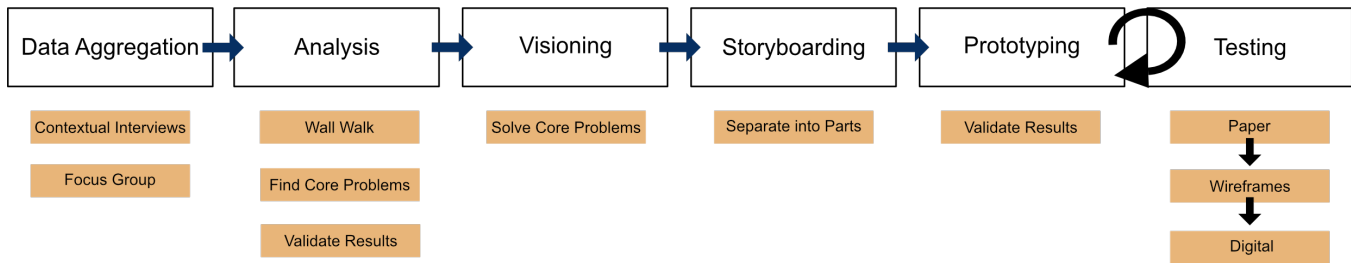


Figure 3: Overview over how we conducted and expanded the complete Rapid Contextual Design process.

storyboards is designed. This prototype is then improved iteratively through *user testing*. We illustrated this process in 3. We had contact persons in the internal medicine, radiology, and oncology stations, whom we could draw on as participants during the whole process. They helped us organize interviews and prototype tests and answered questions concerning request management.

3 DATA AGGREGATION & ANALYSIS

Before starting with our analysis, we received diagrams detailing the current request management process and the staff involved. However, Rapid Contextual Design [5] relies on seeing the process from the user's perspective. For this, we conducted Contextual Interviews, which are unstructured one-on-one interviews in which a user is shadowed by a researcher while doing their everyday work. Meanwhile, the interviewer can ask questions to understand the work in context. The interviewer writes down every piece of information and takes pictures of relevant computer interfaces, the working environment, and everything else of note. As we were not allowed to take photos with names or other identifiable attributes, as patient data is sensitive, we instead created sketches of these artifacts. The unstructured form of those interviews allowed for flexible scheduling and spontaneous changes, e.g., emergencies, without completely derailing the interviews. The short duration of these interviews was a good fit because it allowed us to learn something new while not being overly exhausting for the participants. This gave us the opportunity to watch the everyday workflows and the inevitable workarounds.

We chose to interview twenty-one people drawn from every role to gather data from different perspectives. These interviews were then analyzed soon after so that the interview was still fresh in the

interviewer's mind. The interviewer met with other stakeholders to review the interview for this analysis. Important data was written down on color-coded post-it notes in first-person from the user's perspective. We added the role of the users to the notes so that we could track which aspects are essential for which role. These notes and the sketches we drew during the interviews were then aggregated into a hierarchical collection of data called "*Affinity Diagram*". After the first interview we started the aggregation and added data after each subsequent interview. We created a persona for each role and added them to the Affinity Diagram to help us understand and analyze the data.

After completing and analyzing the interviews, we faced some open questions concerning procedures affecting several roles. As just analyzing each role's responsibilities on their own did not suffice, we conducted an additional focus group with five users (one for each role). This focus group was used to walk through the request management as a group, role by role. We recorded the focus group on video and in a written protocol, which we processed in the same way we analyzed the interviews. We proceeded with adding this data to the Affinity Diagram. The observation of the users and their interactions helped us validate our understanding of the whole process in completion.

Finally, we analyzed the diagram in a *Wall Walk* [5]. In a Wall Walk, all project stakeholders are invited to look over and analyze the Affinity Diagram and add their comments to the data. We identified problems in the current process and brainstormed solutions for each problem together from these comments. As participants from every role were present at this Wall Walk, we were confident that a solution that satisfied everyone had a good chance of success. In total, we found nine problem areas and possible solutions for

Problem area	Severity	Potential Solutions
(1) Difficulties to access patient data & requests	High	1. Aggregate information in easy to find places. 2. Provide mobile requests & patient data.
(2) Large number of phone calls	High	1. Support asynchronous communication channels. 2. Automate examination scheduling.
(3) Restricted & abused access rights	High	1. Provide restricted access to student interns. 2. Let student interns create supervised requests.
(4) Request status difficult to track	High	1. Show current request status. 2. Notify users about important status changes.
(5) Paper notes used for patient data	Medium	1. Provide mobile requests & patient data. 2. Use speech to create requests on the go.
(6) Lack of assistance for data entry	Medium	1. Add designated fields for specific information. 2. Auto-complete requests with patient data.
(7) Frustration through documentation	Medium	1. Provide mobile requests & patient data. 2. Use speech to create requests on the go.
(8) IT-Systems not self-explanatory	Low	1. Add explanatory symbols & tool-tips. 2. Design interfaces with usability in mind.
(9) Conflict between physicians and radiologists	Low	1. Support asynchronous communication channels. 2. Allow changes rather than sending new requests.

Table 1: The problem areas with their severity and possible solutions as published in Krop et al. [11].

them, as summarized in Table 1. Our previous work offers a more detailed version of the found problems and solutions [11].

4 VISIONING & STORYBOARDING

In our analysis, we found nine problem areas and presented potential solutions. First, we changed the request management workflow to better satisfy user needs, as shown on the right side of figure 2. Then, we created a visual illustration of this new workflow, called vision, with the stakeholders. We realized that a mobile solution would not fit everyone, as interns and medical assistants have no iPhones, and radiologists require two monitors to assess radiological images properly. Thus, we decided to design user interfaces for both desktop and mobile applications, with the former having more functions that are not feasible on mobile phones, e.g., scheduling and radiological image assessment.

After every stakeholder was satisfied, we split our two visions into smaller parts detailing specific sub-workflows, e.g., requesting an examination, which we again validated with the stakeholders. Using these, we created storyboards, which describe a contained part of the request management process, e.g., requesting an examination, in more detail. They show how users interact with the system and contain rough sketches of the user interfaces. Once again, we collected the stakeholders' feedback before moving on to prototyping. An example of a storyboard is shown in Figure 4.

5 PROTOTYPING & TESTING

After confirming the Storyboards with the stakeholders, we started designing interface prototypes. We followed an iterative design approach and created paper, wireframe, and digital versions of

the prototype. The paper prototype was drawn by hand, as it can be easily modified during testing and users don't shy away from drawing in solutions. We improved it according to user feedback and designed a wireframe prototype, which was printed instead of drawn and can still be modified easily. Finally, we created a digital prototype using Axure RP 10 [7], which was clickable and reacted to user input, making it close to a final product.

We tested each version with the stakeholders to see if our designs were suitable for request management and how we could improve them. For these tests, each participant received tasks to solve with the prototype. We chose the tasks according to their role in the request management process. For every user, two researchers were conducting the tests. One was protocolling user feedback, while the other led the user through the tasks and (if necessary) controlled the prototype. We used *Simplified Thinking Aloud* [15], where users comment on their every thought about the prototype. This data was then used to improve the prototype and create the next version. After completing their tasks, the participants filled out some questionnaires: We measured usability with the *System Usability Scale* [3], which results in a score between 0 and 100, with a higher score indicating a better usability. The score is divided into five ranges from awful to excellent usability. We also measured task load with the short version of the *NASA Task Load Index* [9]. This scale also results in a score between 0 and 100, with a higher score indicating a higher task load. We gave each participant a short questionnaire about their age and gender to gather demographic data. Additionally, we asked some open questions: *What are the three most essential aspects of request management? What did you like about the prototype? What did you dislike about the prototype?*

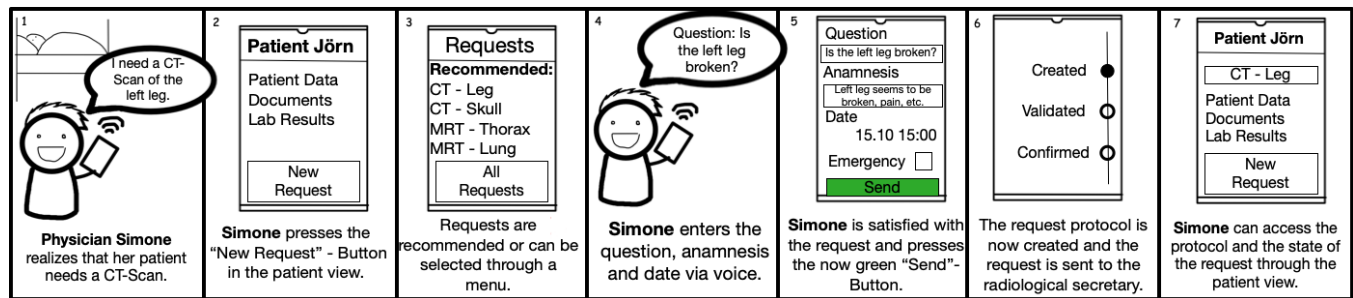


Figure 4: A part of a storyboard of creating a new radiological request.

5.1 Mobile Prototype

Shortly before the project started, the *Service Centre for Medical Informatics* of the *University Clinic of Würzburg* outfitted every physician with an iPhone XR. Thus, we based our initial design on Apple's *Human Interface Guidelines for iOS* [1], as we suspected it might ease the traversal to our application and similar approaches using Google's *Material Design* have been used before [14]. The first version of the prototype mainly focused on the tasks of the request management process that suited mobile devices, such as issuing new requests, reviewing by radiologists, and the communication between physicians, radiologists, and medical assistants: We added a single place to access all patient documents and requests accessible from the station list. We revised the request form by removing unnecessary fields and automatically attaching known risk factors, like allergies, to the request. Each request now contains a chat protocol accessible by each person working on an examination. Status changes are reflected in this chat and in a timeline, allowing the user to see the request status at a glance. Since the prototype is intended for mobile devices, we also added an opt-in system for notifications whenever the status changes.

In total, 46 participants ($M_{age} = 31.98$, $SD_{age} = 5.38$) tested the system. We tried to recruit participants with no prior experience with the prototype for each test. Still, as we relied on a contact person in each station and had no final say over the recruitment process, six participants were tested twice. We still included these participants in our analysis since the prototype changed drastically between each test round and the interactions when switching materials, i.e., a paper prototype feels and works distinct from a digital one. Twelve participants (five female, $M_{age} = 34.80$, $SD_{age} = 8.21$) tested the paper version. This prototype achieved good usability ($M = 81.87$, $SD = 13.66$) and a low task load ($M = 21.66$, $SD = 13.2$). Fifteen (nine female, $M_{age} = 31.10$, $SD_{age} = 4.07$) tested the wireframes. The wireframes achieved slightly better usability ($M = 84$, $SD = 12.70$) and lower task load ($M = 17.27$, $SD = 12.49$) than the paper prototype. Our participants praised the design as clean and intuitive, except one who did not like the color. We deployed the digital prototype with the *Axure RP* application on an iPhone XR, which allowed us to conduct these user tests very close to reality. For this version, we tested nineteen participants (seven female, $M_{age} = 31.80$, $SD_{age} = 5.22$). It achieved the highest usability score of the three prototypes ($M = 87.5$, $SD = 9.20$) and a low task load ($M = 18.64$, $SD = 13.68$). How the usability and task load changed over time can be seen in Figure 5.

The qualitative feedback for each testing round was overwhelmingly positive and indicated that our prototype was moving in the right direction concerning design and functionality. This feedback also showed that *fast and intuitive use*, *fast processing of the request in general*, and the *focus on the quality of the request* are perceived as crucial facets to improve request management. These also were the things participants liked most about our prototype, as they especially liked that the prototype was *fast and intuitive to use*, that the *design was clean and consistent* and that the *request process was fast and easy*. They also liked that we improved communication channels and made important information *more readily available*, e.g., risk factors. On the contrary, participants *were missing some interactions*, like seeing risk factors during planning, and *found some interactions non-intuitive*, like having an extra checkbox to indicate emergencies. We used this feedback to improve our prototype accordingly and added missing functionalities.

5.2 Desktop Prototype

We realized that a mobile solution would not cover all stakeholders and their needs during the visioning since radiologists need two monitors to assess radiological images correctly, and medical assistants did not receive an iPhone. Thus, we decided to design a user interface for a desktop application, which application contains all features of the mobile solution and offers additional workflows only feasible on desktop PCs, e.g., scheduling and radiological image assessment. We decided to test it as paper, wireframe, and digital prototype versions as well. Our initial mock-ups and the digital version can be seen in Figure 1.

In total, 44 participants (22 female, $M_{age} = 31.11$, $SD_{age} = 6.64$) took part in our tests for the desktop application. We tried to recruit users who had not seen any of our prototypes before, as we wanted them to experience it as unbiased as possible. However, as we only had access to three cooperating stations and tested most of the medical staff for the mobile application already, we had to retest seven participants. We tested the paper version with three radiologists (one female, two male, $M_{age} = 28.67$, $SD_{age} = 1.15$). Again, we used the SUS to measure usability and the TLX-R to determine task load. The prototype achieved good usability ($M = 70.00$, $SD = 18.87$) and caused a medium task load ($M = 31.67$, $SD = 15.90$). Before we could test participants from other stations, the global COVID-19 pandemic started to affect the country. As paper prototypes have to be touched and users and the testers are sitting close, it was impossible to conduct tests while keeping hygiene rules.

Thus, we decided to skip the remaining tests for the paper prototype, as we could not replicate paper prototyping to a satisfying degree in remote testing sessions.

Instead, we started testing the wireframes remotely using Miro boards [13] and Zoom [8]. To simulate the prototype, we placed interface elements that the tester could move around. The user was zoomed in on a part of the screen so that it felt like the prototype filled out their whole screen. The user's mouse movement was visible to others on the same Miro board. Besides that, everything else in the testing process remained the same. After finishing all Use-Cases, we asked participants to disconnect from Zoom and fill out the questionnaires. Even though the first few tests with this remote solution felt weird, it worked surprisingly well. The resonance to remote testing was positive, and every user understood the process after a short explanation. We faced initial technical problems, like the clinic PCs only supporting older Windows Versions or having no microphones. However, it was also easier to recruit physicians, as more of them reacted to our mails or could also do these tests from home. This was a positive outcome that we did not expect. The reasons for why more users participated are unclear, but as many appeared in Zoom with their camera turned off, we suspect that it lowered the barrier of meeting new people. Also, we suspect that physicians paid more attention to their mails, as more aspects of life have shifted to online. The remote testing sessions were also shorter compared to in-person testing. This was partly because the participants filled out the questionnaires without us but also because we had the feeling that participants were more focused. Compared to in-person tests, we had hardly any small talk in remote testing sessions.

In total, twenty-five participants (ten female, $M_{age} = 31.16$, $SD_{age} = 6.84$) took part in the remote wireframe tests. This prototype achieved excellent usability ($M = 85.20$, $SD = 18.37$) and caused a low task load ($M = 14.77$, $SD = 9.92$). We quickly finished these tests, improved the prototype according to feedback, and continued with the digital version, which we again tested remotely using Axure RP 10 and Zoom. We tested sixteen participants (11 female, $M_{age} = 31.5$, $SD_{age} = 7.06$). This version achieved even better usability ($M = 89.69$, $SD = 12.58$) and lower task load ($M = 12.97$, $SD = 12.68$) than the previous versions. It was well received, and multiple physicians asked us when they could finally replace their current user interfaces with ours. With each iteration, the usability improved while the task load decreased. These developments can be seen in Figure 5.

The qualitative feedback from the open questions was immensely positive. As with the mobile prototype, Participants liked the *fast and intuitive use*, the *consistent and clear design*, and the *simple and easy request management*. However, they deemed *easy access to relevant information* more important than *fast and intuitive use*. Again, we used negative feedback to improve the prototype. Participants did not like that the prototype *lacked some options*, which were outside this project's scope. Additionally, some did not like the *design and the dominant use of blue color*, which we chose to be consistent with the mobile version and the clinic's corporate design. We also improved many minor aspects, like how information is shown about handling infectious patients. After finishing our tests, we sent the official specification to the Service Centre of

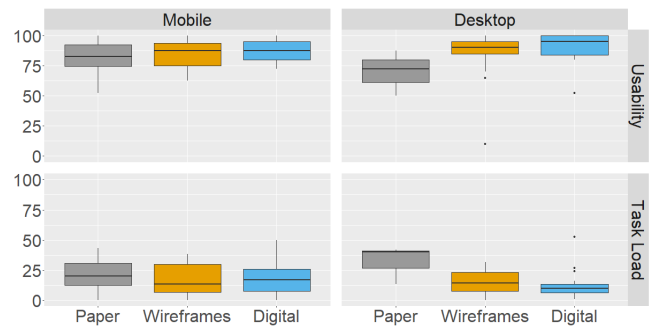


Figure 5: The development of the usability (System Usability Scale, top) and task load (NASA-TLX-Raw, bottom) of the mobile and desktop prototypes. The final prototypes achieved excellent usability scores and caused a low task load.

Medical Informatics, but, as of writing, they have not implemented the system yet.

6 DISCUSSION AND IMPLICATIONS

Overall, the *Rapid Contextual Design* approach was well-fitting to develop a new workflow and a supporting interface in environments where users with multiple roles are involved. Although we deviated from the process, e.g., by conducting an additional Focus Group, the users felt included in the design process. The hospital setting proved to be a challenging environment for User-Centered Design, as users are hard to reach and have hard time constraints. We cooperated only with some stations of the university hospital. It was not feasible to replace existing software solutions since other stations still rely on the underlying database, which is needed for sharing documents and communication between stations. Thus, we tried to solve the problems found in our analysis while still being compatible with underlying databases. Even though our participants were enthusiastic about the new workflow, it was challenging to schedule meetings. Our contact person in each participating station arranged the communication by providing us with mail addresses, but sometimes mails were forgotten or left unanswered. We tried scheduling through software but found that even fewer physicians responded through it than through emails. Thus, we resumed communication through mail, which got better after the start of the pandemic. As these tests occurred in a hospital setting with working physicians, interruptions were frequent since participants often received urgent notifications or phone calls. Sometimes, participants arrived late to scheduled meetings, as an emergency occurred shortly before the meeting started. At first, we thought they had forgotten us, but after this happened a few times, we came to expect this behavior and told participants that this was completely fine.

The test results of our prototypes were very satisfying. Qualitative and quantitative feedback indicates that we are on the right path with our workflow and design changes. In the beginning, many participants suggested improvements and mentioned some things we forgot while designing the paper prototypes. Fortunately, we were able to satisfyingly integrate these improvements in later iterations, as they did not come up again, even when we asked directly. Many

participants mentioned how much they wished they could use our system immediately instead of the old one and now impatiently expect its implementation. The quantitative data gave us insight into the progression of our designs, while the qualitative data was helpful to improve it further. The perceived usability increased for both the mobile and desktop prototypes while the task load simultaneously decreased. The differences between iterations were not significantly different. We attribute this to a) prototype testing not being a valid scenario to perform significance tests and b) the low participant count in each prototype round. However, it gave us a rough indication that we were on the right track, supported by our qualitative results.

Due to the global COVID-19 pandemic, we had to switch to remote user testing midway through the process. Initially, this presented some technical challenges concerning hardware and software. After our partners prepared one PC per station with the necessary tools, we could test the users one after another. Some participants, especially older physicians, lacked technical knowledge and had problems connecting to Zoom. Even so, with remote help, detailed explanations, and lots of patience, we managed to test the prototypes with every suitable candidate. In two cases, screen sharing was impossible from the participant's side due to slow connection speeds. In a spontaneous workaround, one of the authors shared their screen with the participant indicating where they wanted to click next. All in all, this workaround worked well and led to results comparable to all other tests.

Remote user testing had its advantages as well: First, it is more hygienic due to participants not needing to touch the prototype by hand and the spatial separation by being remote. Second, we could recruit participants more easily than when we tested in-house. The tests were shorter in general, but due to us being more flexible with our time slots, more participants were able to participate in the study. Third, we had the feeling that participants felt more motivated and took us more seriously after we started to use sophisticated tools like *Miro*, *Zoom*, and *Axure RP* instead of when we arrived with a paper prototype. These advantages make remote prototype testing attractive in settings where time allocation is complex or users are hard to reach. This makes it a good alternative to in-person testing, even despite the technical challenges.

Overall, we were satisfied with how well analyzing this complex and distributed process worked, and we are confident that our interfaces support the new workflow. Supplementing Rapid Contextual Design with a Focus Group was a major improvement since it helped us understand the interconnections between each role. We were also surprised how well switching to remote user tests turned out. Not only did we get good results, we also learned what works and what doesn't when conducting remote user testing in hospitals. Based on our experiences, we propose five best practices for using *Rapid Contextual Design* in hospitals:

- (1) **Supplementing Rapid Contextual Design with focus groups is well suited to identify problems and solutions in a complex process with multiple roles.**
- (2) **Have a contact person for each relevant role to better understand their specific tasks.**
- (3) **Be prepared for interrupts and delays in the testing sessions.**

- (4) **Prepare a workstation with all necessary tools where users can participate in your tests.**
- (5) **Remote prototype testing works as well as in-person testing. It is helpful if users are hard to reach or scheduling is complicated.**

7 CONCLUSION

This paper reported a user-centered design approach for IT-supported optimization of a radiological request workflow. It is based on *Rapid Contextual Design* and is a part of a larger interdisciplinary project within our partner clinic. We performed unstructured interviews and aggregated the data in an Affinity Diagram. Based on these results, we identified nine core problems and created a vision and storyboards for a new workflow to solve them. Then, we iteratively developed prototypical user interfaces for a mobile and desktop application, which support the new workflow. Each prototype iteration was usability-tested. The evaluations resulted in excellent usability scores while also indicating low task loads. Our qualitative and quantitative results indicate that we chose the right approach and that our solution works as intended.

In addition, the chosen methods to tackle the overall problem were well-suited to achieve our goals. The involvement of users from multiple roles and stakeholders led to a holistic approach, where we analyzed current workflows and solved their problems. We recommend adding a Focus Group to every requirement analysis where multiple roles are involved, as it significantly improved our understanding of the whole process. Remote prototype testing yielded good quantitative and qualitative results and eased scheduling and organization. Albeit we faced initial difficulties, the combination of Zoom, Miro, and Axure RP for remote prototyping worked very well. We formulated the outcome of our experiences as five best practices, which can easily be adapted when switching to remote prototyping.

We are now awaiting the implementation of the final prototype by the Service Center Medical Informatics of the university hospital. In the future, we would like to compare this implementation to the current way to request and manage procedures used in the hospital. We suspect that our solution might reduce time and resources spent on request management and lead to increased time for patient care and increased request quality instead.

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